Smoother river bars

Continuous exposed bars and better shallow-water geometry

River bars now blend continuously along connected cross-sections. At the tested problem reach, a 4.8-metre bed jump across ten centimetres of travel fell to about three centimetres. Inner bends retain broader exposed sediment; outer banks stay narrower and more vegetated.



Earlier overview of river reach 115180.



Revised bars on the same reach. The selected anchor and camera differ; this is not an aligned pixel comparison.



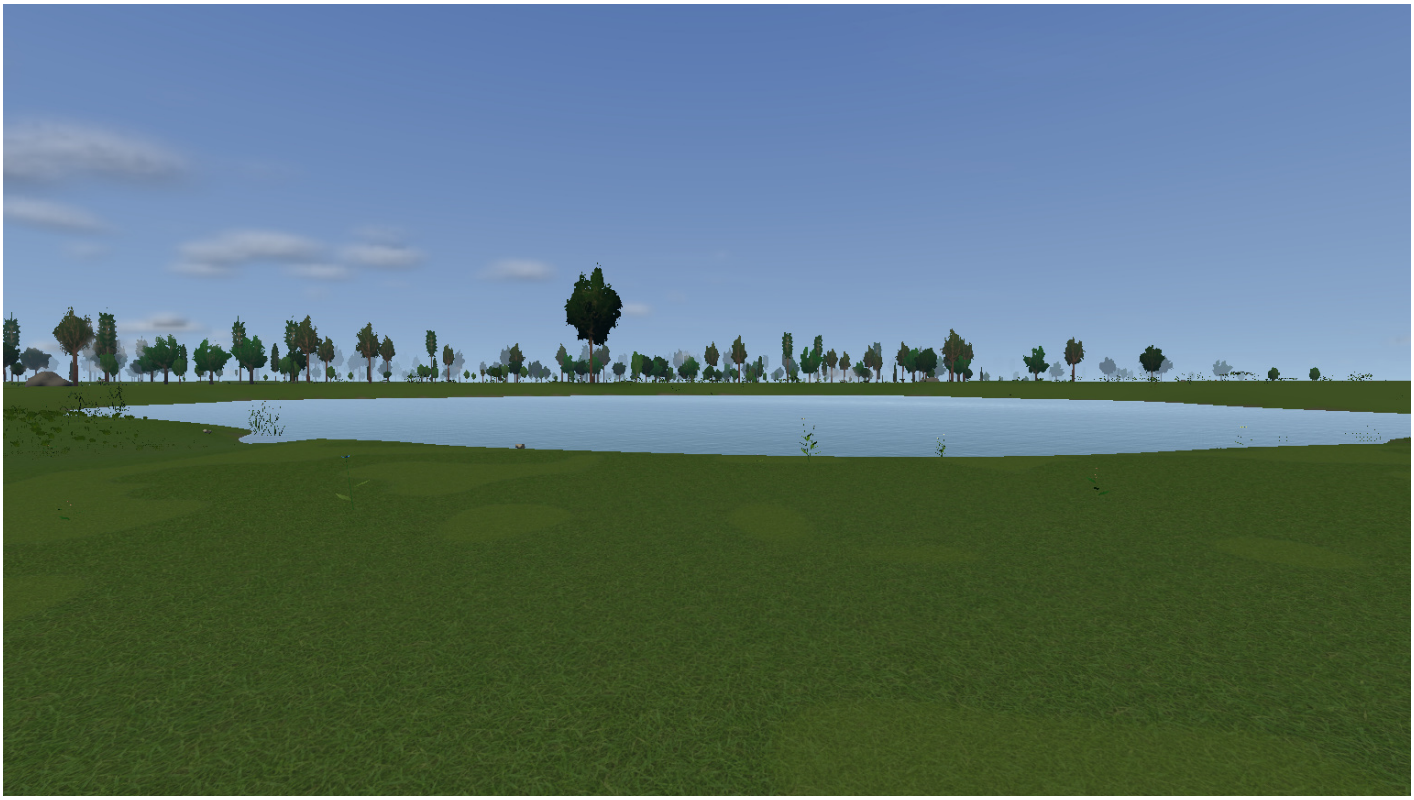
The revised exposed bar from ground level. Existing generated ground textures are retained.

Some courses remain too ribbon-like, and a separate legacy bank-overlap defect remains unresolved.

Finite ponds, continuous shores

Small water bodies stay inside their actual basins

Coarse ground triangles could bridge a pond hollow while water disappeared beneath them, or extend wet ground beyond its finite shore. Encountered ponds now receive bounded basin refinement, and the water follows the same curved datum as the terrain.



Actual in-game shoreline review. This near view is unchanged by the later reduction in distant-only refinement.

Bounded cost

Refinement applies to finite ponds that need it. Distant ponds use a smaller constraint pattern. Terrain, water, vegetation and saved-history checks accompany the visual review.

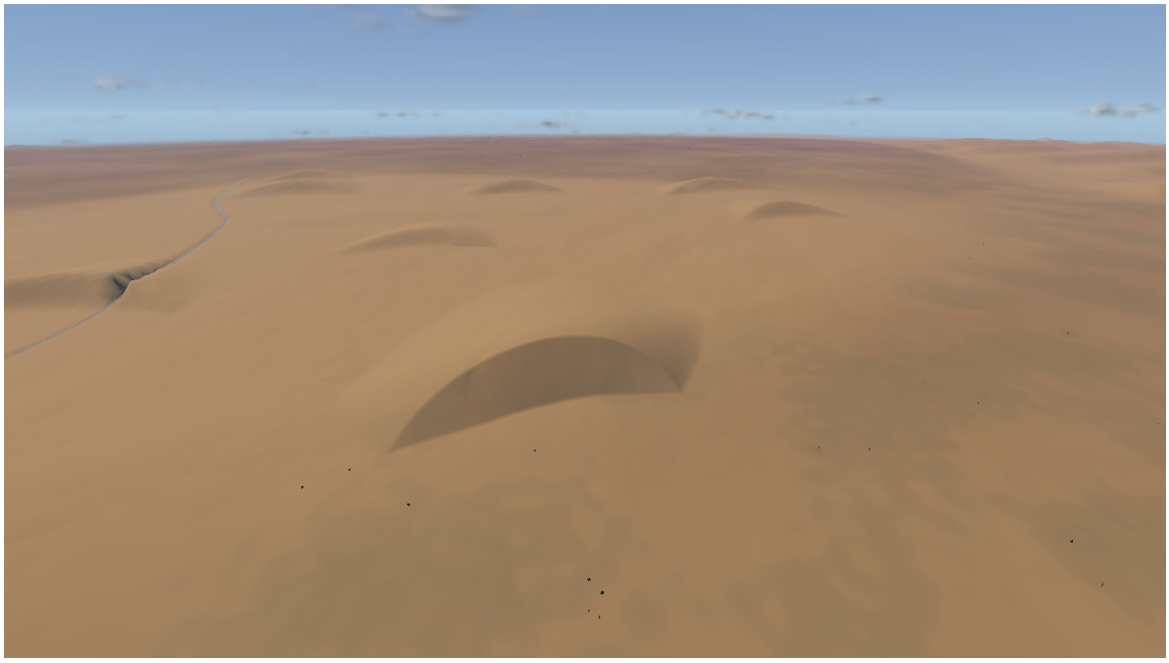
Still to improve

Vegetation placement and art need further work. Shallow-water planting is shown on page 4.

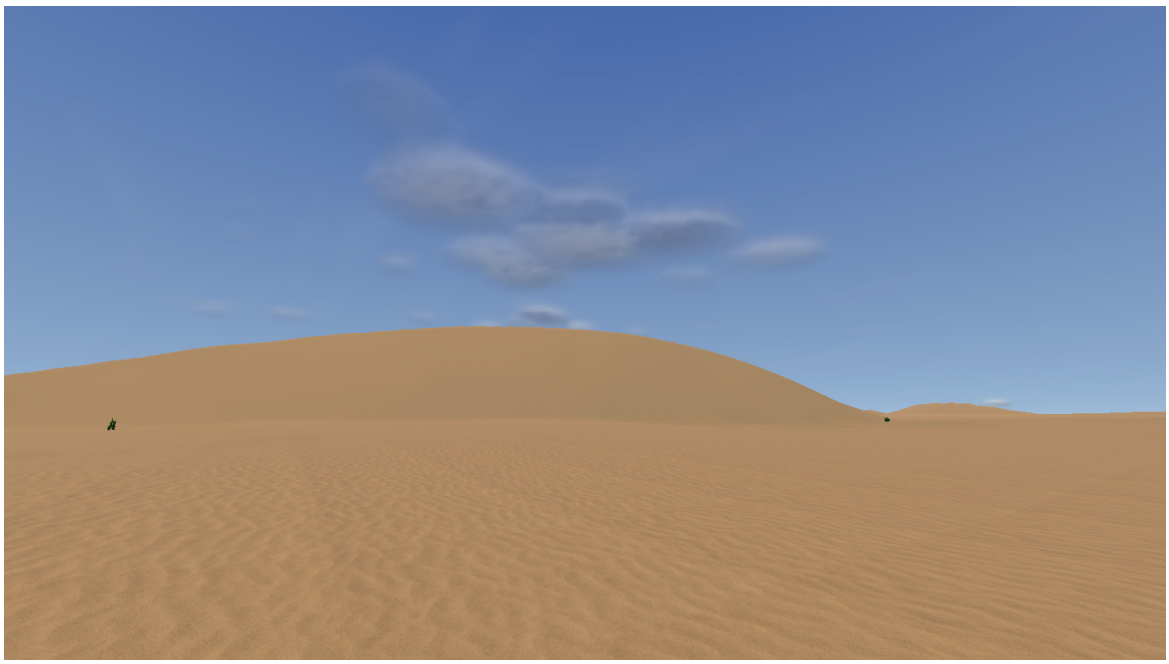
Crescent dune fields

A distinct sand landscape alongside the existing connected ridges

Separated crescent dunes add curved crests, tapered horns and a steeper lee face. Their placement uses existing sandy terrain, seasonal wind diversity and a bounded terrain-suitability proxy.



An actual generated crescent field, viewed obliquely. Medium vegetation settings.



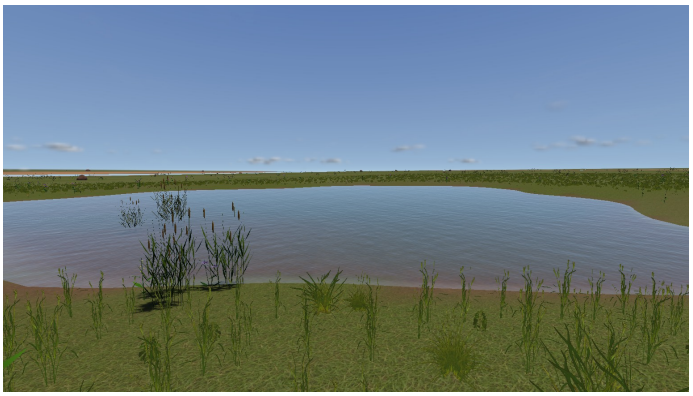
Ground-level view of the dune face. No extra scenery objects or new rendering passes were added.

Wetlands and tidal planting

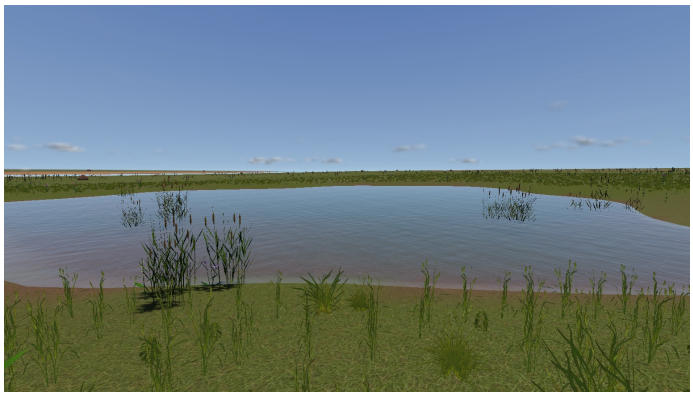
Plants rooted in shallow water, with coastal grasses/rushes

Included in the updated build selected by the usual WorldGen shortcut.

Wetland plants can now grow in shallow water and below the old two-metre elevation cutoff. Coastal variants use tidal habitat and salt tolerance, with grass/rush shapes. Depth, current, temperature, soil and canopy limits remain.



Earlier freshwater shoreline.



More emergent stands in the same freshwater setting.



Earlier mangrove waterline.



Coastal emergents at the same camera and waterline.

A twofold establishment boost adds emergent stands; distant reeds use simple shared meshes. Checked terrain and water geometry match the prior version. Vegetation art still needs refinement.

Both before/after pairs use identical camera files and actual game captures.

Checks and remaining work

Matched moving routes and checks on the selected build

RTX 4060 / Ryzen 7 5700X; 1920 x 1080; Medium; 6 km view; clear local-summer noon. Each route runs a 45-second, 180-metre loop with rapid turns. Timing traces were enabled. The owner confirmed the PC idle. All 167,776 recorded frames were focused.

Route	Before FPS	Updated FPS	Updated p99 / max
Crescent dunes	775	783	2.45 / 12.14 ms
Dense jungle	204	205	10.15 / 17.44 ms
Mangrove waterline	413	391	5.67 / 11.70 ms
Salt marsh	494	462	4.88 / 30.35 ms

Dry routes are effectively unchanged. Added wetland vegetation costs about 5-6% at these sites. Streaming remained active, and scene contents differ deliberately in wetlands. These four routes do not establish a global worst case or a guaranteed frame floor.

Additional verified fixes

Coarse terrain/map queries previously depended on earlier queries, differing by up to 2.43 metres. The correction passed 810 order checks, with 21 complete game payload comparisons unchanged.

Proven zero-forest nodes now skip empty placement work. Tested empty L4 requests dropped from 262,144 probes to zero; checked jungle/coastal tree data stayed identical. No total-startup gain is claimed.

Still open

The absolute 100 FPS floor remains unmet. Spikes, a legacy river-bank gap, regular smaller river shapes and wetland art remain follow-ups. The failed river-width experiment is isolated on its own Git branch and is absent from this build.

Trace buffers can allocate during measurement. A final dune check without detailed diagnostics averaged 771 FPS, with an 11.14 ms worst frame; removing diagnostics did not eliminate every spike. Their normal-game causes remain unresolved.

Research: NPS - meandering streams; NPS - dune types. Captures and PDF rebuild inputs are preserved beside this report; verification records are under assets/verification.